

INFORMATION THEORY & CODING

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Review Summary

- **Definition** (*Entropy*) The *entropy* $H(X)$ of a discrete random variable X is defined by

$$H(X) = - \sum_{x \in \mathcal{X}} p(x) \log p(x).$$

- **Properties of H**

1. $H(X) \geq 0$
2. $H_b(X) = (\log_b a) H_a(X)$

Review Summary

- **Definition** (*Relative entropy*) The *relative entropy* $D(p||q)$ of the probability mass function p with respect to the probability mass function q is defined by

$$D(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

- **Definition** (*Mutual information*) The *mutual information* between two random variables X and Y is defined as

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

Review Summary

■ Alternative expressions and relations

$$H(X) = E_p \log \frac{1}{p(X)}$$

$$H(X, Y) = E_p \log \frac{1}{p(X, Y)}$$

$$H(X|Y) = E_p \log \frac{1}{p(X|Y)}$$

$$I(X; Y) = E_p \log \frac{p(X, Y)}{p(X)p(Y)}$$

$$D(p||q) = E_p \log \frac{p(X)}{q(X)}$$

$$\begin{aligned} I(X; Y) &= H(X) - H(X|Y) \\ &= H(Y) - H(Y|X) \\ &= H(X) + H(Y) - H(X, Y). \end{aligned}$$

Review Summary

■ Chain rules

Entropy:

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1).$$

Mutual information:

$$I(X_1, X_2, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y | X_1, X_2, \dots, X_{i-1}).$$

Relative entropy:

$$D(p(x, y) || q(x, y)) = D(p(x) || q(x)) + D(p(y|x) || q(y|x)).$$

Convexity & Concavity

- **Definition.** (*Convexity*) A function $f(x)$ is said to be *convex* over an interval (a, b) if for every $x_1, x_2 \in (a, b)$ and $0 \leq \lambda \leq 1$,

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2).$$

A function f is called *strictly convex* if equality holds *only if* $\lambda = 0$ or $\lambda = 1$.

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- **Definition.** (*Concavity*) A function f is *concave* if $-f$ is convex. A function is convex if it always lies below any chord. A function is concave if it always lies above any chord.

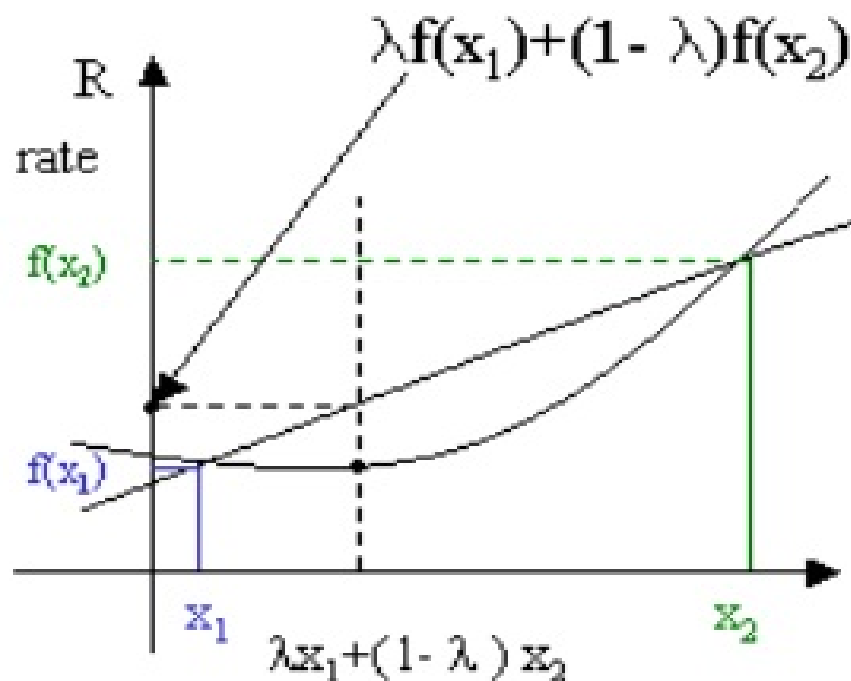
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- **Definition.** (*Concavity*) A function is *concave* if it is convex. A function is *strictly concave* if it is strictly convex. A function is *convex* if it is not strictly concave.



A Theorem

■ Examples

$$f(x) = x^2, \quad |x|, \quad e^x, \quad x \log x \quad (x > 0)$$

$$g(x) = \log x, \quad \sqrt{x}, \quad (x \geq 0)$$

注意：中国大陆数学界某些机构关于函数凹凸性定义和国外的定义是相反的。Convex Function在某些中国大陆的数学书中指凹函数。Concave Function指凸函数。但在中国大陆涉及经济学的很多书中，凹凸性的提法和其他国家的提法是一致的，也就是和数学教材是反的。举个例子，同济大学高等数学教材对函数的凹凸性定义与本条目相反，本条目的凹凸性是指其上方图是凹集或凸集，而同济大学高等数学教材则是指其下方图是凹集或凸集，两者定义正好相反。

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Theorem If the function f has a **second derivative** that is **nonnegative (positive)** over an interval, the function is **convex (strictly convex)** over that interval.

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Theorem If the function f has a **second derivative** that is **nonnegative (positive)** over an interval, the function is **convex (strictly convex)** over that interval.

Proof.

Jensen's Inequality

- **Theorem 2.6.2** (*Jensen's inequality*) If f is a **convex** function and X is a random variable,

$$Ef(X) \geq f(EX).$$

Moreover, if f is **strictly convex**, the equality above implies that $X = EX$ with probability 1 (i.e., **X is a constant**).

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Proof. **By mathematical induction.**

- $k = 2$:

$$p(x_1)f(x_1) + p(x_2)f(x_2) \geq f(p(x_1)x_1 + p(x_2)x_2).$$

- Hypothesis: $\sum_{i=1}^{k-1} p(x_i)f(x_i) \geq f\left(\sum_{i=1}^{k-1} p(x_i)x_i\right).$
- Induction: $\sum_{i=1}^k p(x_i)f(x_i).$

Information Inequality

- **Theorem 2.6.3** (*Information inequality*) Let $p(x), q(x), x \in \mathcal{X}$, be two probability mass functions. Then

$$D(p||q) \geq 0$$

with equality **iff** $p(x) = q(x)$ for all x .

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with equality iff $p(x) = q(x)$ for all x .

Proof. Let $A = \{x : p(x) > 0\}$ be the support set of $p(x)$. Then

$$\begin{aligned} -D(p||q) &= -\sum_{x \in A} p(x) \log \frac{p(x)}{q(x)} \\ &= \sum_{x \in A} p(x) \log \frac{q(x)}{p(x)} \\ &\leq \log \sum_{x \in A} p(x) \frac{q(x)}{p(x)} \\ &= \log \sum_{x \in A} q(x) \\ &\leq \log \sum_{x \in \mathcal{X}} q(x) \\ &= 0 \end{aligned}$$

Corollaries

- **Corollary** (*Nonnegativity of mutual information*) For any two random variables X and Y ,

$$I(X; Y) \geq 0,$$

with equality iff X and Y are independent.

- **Corollary**

$$D(p(y|x)||q(y|x)) \geq 0,$$

with equality iff $p(y|x) = q(y|x)$ for all y and x such that $p(x) > 0$.

- **Corollary**

$$I(X; Y|Z) \geq 0,$$

with equality iff X and Y are conditionally independent given Z .

The maximum entropy distribution

- **Theorem 2.6.4** $H(X) \leq \log |\mathcal{X}|$, where $|\mathcal{X}|$ denotes the number of elements in the range of X , with equality **iff** X has a uniform distribution over $\mathcal{X}$.

The maximum entropy distribution

- **Theorem 2.6.4** $H(X) \leq \log |\mathcal{X}|$, where $|\mathcal{X}|$ denotes the number of elements in the range of X , with equality **iff** X has a uniform distribution over $\mathcal{X}$.

Proof Let $u(x) = \frac{1}{|\mathcal{X}|}$ be the uniform probability mass function over $\mathcal{X}$, and let $p(x)$ be the probability mass function for X . Then

$$0 \leq D(p||u) = \sum p(x) \log \frac{p(x)}{u(x)} = \log |\mathcal{X}| - H(X).$$

Conditioning reduces entropy

- **Theorem 2.6.5** (*Conditioning reduces entropy*)

$$H(X|Y) \leq H(X)$$

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- **Example** Let (X, Y) have the following joint distribution:

$\begin{array}{c} X \\ \diagdown \\ Y \end{array}$	1	2
1	0	$\frac{3}{4}$
2	$\frac{1}{8}$	$\frac{1}{8}$

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- **Example** Let (X, Y) have the following joint distribution:

$X \backslash Y$	1	2
1	0	$\frac{3}{4}$
2	$\frac{1}{8}$	$\frac{1}{8}$

$H(X) = 0.544$ bit, $H(X|Y = 1) = 0$ bit, $H(X|Y = 2) = 1$ bit,
 $H(X|Y) = 0.25$ bit

Conditioning reduces entropy

- **Theorem 2.6.5** (*Conditioning reduces entropy*)

$$H(X|Y) \leq H(X)$$

with equality **iff** X and Y are independent.

- **Theorem 2.6.6** (*Independence bound on entropy*) Let $X_1, X_2, \dots, X_n$ be drawn according to $p(x_1, x_2, \dots, x_n)$, Then

$$H(X_1, X_2, \dots, X_n) \leq \sum_{i=1}^n H(X_i)$$

with equality **iff** the X_i 's are independent.

Data-processing inequality

- **Definition.** (*Markov Chain*) Random variables X, Y, Z are said to *form a Markov chain* in that order (denoted by $X \rightarrow Y \rightarrow Z$) if the conditional distribution of Z **depends only on Y** and is conditionally **independent of X** . Specifically, X, Y and Z form a Markov chain $X \rightarrow Y \rightarrow Z$ if the joint probability mass function can be written as

$$p(x, y, z) = p(x)p(y|x)p(z|y).$$

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$$p(x, y, z) = p(x)p(y|x)p(z|y).$$

- $X \rightarrow Y \rightarrow Z \Rightarrow p(x, z|y) = p(x|y)p(z|y)$
- $X \rightarrow Y \rightarrow Z \Rightarrow Z \rightarrow Y \rightarrow X$
- If $Z = f(Y)$, then $X \rightarrow Y \rightarrow Z$.

Data-processing inequality

- **Theorem 2.8.1** (*Data-processing inequality*) If $X \rightarrow Y \rightarrow Z$, then $I(X; Y) \geq I(X; Z)$.

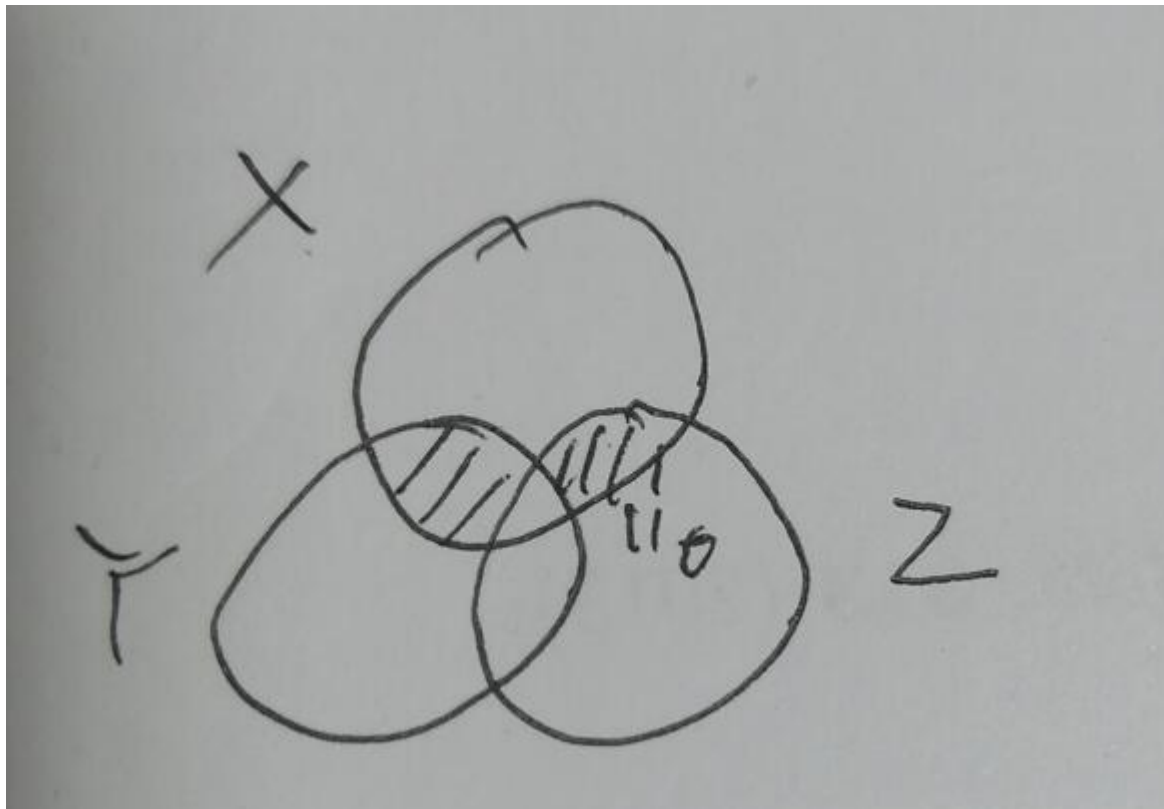
Proof. By the chain rule, we expand $I(X; Y, Z)$ in two ways:

$$\begin{aligned} I(X; Y, Z) &= I(X; Z) + I(X; Y|Z) \\ &= I(X; Y) + I(X; Z|Y). \end{aligned}$$

Since $X \rightarrow Y \rightarrow Z$, we have $I(X; Z|Y) = 0$. Since $I(X; Y|Z) \geq 0$, we have $I(X; Y) \geq I(X; Z)$.

Corollaries

- **Corollary** In particular, if $Z = g(Y)$, we have $I(X; Y) \geq I(X; g(Y))$.
- **Corollary** If $X \rightarrow Y \rightarrow Z$, then $I(X; Y|Z) \leq I(X; Y)$.



Fano's inequality

- **Problem 2.5** (*Zero conditional entropy*) Show that if $H(X|Y) = 0$, then X is a function of Y , i.e., for all y with $p(y) > 0$, there is **only one** possible value of x with $p(x, y) > 0$.

Fano's inequality

- **Proposition 2.5** (*Zero conditional entropy*) Show that if $H(X|Y) = 0$, then X is a function of Y , i.e., for all y with $p(y) > 0$, there is **only one** possible value of x with $p(x, y) > 0$.

Proof. Assume that there exists an y , say y_0 and two different values of x , say x_1 and x_2 such that $p(y_0, x_1) > 0$ and $p(y_0, x_2) > 0$. Then $p(y_0) \geq p(y_0, x_1) + p(y_0, x_2) > 0$, and $p(x_1|y_0)$ and $p(x_2|y_0)$ are not equal to 0 or 1. Thus,

$$\begin{aligned} H(X|Y) &= - \sum_y p(y) \sum_x p(x|y) \log p(x|y) \\ &\geq p(y_0) (-p(x_1|y_0) \log p(x_1|y_0) - p(x_2|y_0) \log p(x_2|y_0)) \\ &> 0, \end{aligned}$$

since $-t \log t \geq 0$ for $0 \leq t \leq 1$, and is strictly positive for $t \neq 0, 1$, which is a contradiction to $H(X|Y) = 0$.

Fano's inequality

- The conditional entropy of a random variable X given another random variable Y is zero ($H(X|Y) = 0$) iff X is a function of Y . Hence we can estimate X from Y with zero probability of error iff $H(X|Y) = 0$.

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- We can estimate X with a low probability of error P_e only if the conditional entropy $H(X|Y)$ is small. *Fano's inequality* quantifies this idea.

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Why do we need to related P_e to entropy $H(X|Y)$?

Fano's inequality

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Why do we need to related P_e to entropy $H(X|Y)$?

When we have a communication system, we send X , but receive a corrupted version Y . We want to infer X from Y . Our estimate is $\hat{X}$ and we will make a mistake as

$$P_e = \Pr[\hat{X} \neq X]$$

Markov chain $X \rightarrow Y \rightarrow \hat{X}$.

Fano's inequality

- **Problem.** A random variable Y is related to another random variable X with a distribution $p(x)$. From Y , we calculate a function $g(Y) = \hat{X}$, where $\hat{X}$ is an estimate of X and takes on values in $\hat{\mathcal{X}}$. We observe that $X \rightarrow Y \rightarrow \hat{X}$ forms a Markov chain. How to bound the estimate error probability $P_e = \Pr[\hat{X} \neq X]$?

Fano's inequality

- **Theorem 2.10.1** (*Fano's Inequality*) For any estimator $\hat{X}$ such that $X \rightarrow Y \rightarrow \hat{X}$, with $P_e = \Pr\{X \neq \hat{X}\}$, we have

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \geq H(X|\hat{X}) \geq H(X|Y).$$

This inequality can be weakened to

$$1 + P_e \log(|\mathcal{X}| - 1) \geq H(X|Y)$$

or

$$P_e \geq \frac{H(X|Y) - 1}{\log |\mathcal{X}| - 1}.$$

Proof.

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Proof. Define an error random variable as

$$E = \begin{cases} 1 & \text{if } \hat{X} \neq X, \\ 0 & \text{if } \hat{X} = X. \end{cases}$$

Using the chain rule for entropies to expand $H(E, X|\hat{X})$ in two different ways, we have

$$\begin{aligned} H(E, X|\hat{X}) &= H(X|\hat{X}) + \underbrace{H(E|X, \hat{X})}_{=0} \\ &= \underbrace{H(E|\hat{X})}_{\leq H(P_e)} + \underbrace{H(X|E, \hat{X})}_{\leq P_e \log(|\mathcal{X}| - 1)}. \end{aligned}$$

Since *conditioning reduces entropy*, $H(E|\hat{X}) \leq H(E) = H(P_e)$. Since E is a function of X and $\hat{X}$, the conditional entropy $H(E|X, \hat{X})$ is equal to 0. We now look at $H(X|E, \hat{X})$. By the equation $H(X|Y) = \sum_y p(y)H(X|Y = y)$, we have

$$\begin{aligned} H(X|E, \hat{X}) &= \sum_{\hat{x} \in \mathcal{X}} \{ \Pr[\hat{X} = \hat{x}, E = 0] H(X|\hat{X} = \hat{x}, E = 0) \\ &\quad + \Pr[\hat{X} = \hat{x}, E = 1] H(X|\hat{X} = \hat{x}, E = 1) \}. \end{aligned}$$

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By definition of E , X is **conditionally deterministic** given $\hat{X} = \hat{x}$ and $E = 0$, then $H(X|\hat{X} = \hat{x}, E = 0) = 0$. If $\hat{X} = \hat{x}$ and $E = 1$, then X must take a value in the set $\{x \in \mathcal{X} : x \neq \hat{x}\}$ which contains $|\mathcal{X}| - 1$ elements. Then $H(X|\hat{X} = \hat{x}, E = 1) \leq \log(|\mathcal{X}| - 1)$.

$$\begin{aligned} H(X|E, \hat{X}) &\leq \sum_{\hat{x} \in \mathcal{X}} \Pr[\hat{X} = \hat{x}, E = 1] \log(|\mathcal{X}| - 1) \\ &= \Pr[E = 1] \log(|\mathcal{X}| - 1) \\ &= P_e \log(|\mathcal{X}| - 1). \end{aligned}$$

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$$\begin{aligned} H(X|E, \hat{X}) &\leq \sum_{\hat{x} \in \mathcal{X}} \Pr[\hat{X} = \hat{x}, E = 1] \log(|\mathcal{X}| - 1) \\ &= \Pr[E = 1] \log(|\mathcal{X}| - 1) \\ &= P_e \log(|\mathcal{X}| - 1). \end{aligned}$$

By the data-processing inequality, we have $I(X; \hat{X}) \leq I(X; Y)$ and therefore $H(X|\hat{X}) \geq H(X|Y)$.

Corollary

- **Corollary** For any two random variables X and Y , let $p = \Pr(X \neq Y)$.

$$H(p) + p \log(|\mathcal{X}| - 1) \geq H(X|Y).$$

Proof. Let $\hat{X} = Y$ in Fano's inequality.

Fano's inequality

- **Remark** Suppose that there is **no knowledge** of Y . Thus, X must be guessed without any information. Let $X \in \{1, 2, \dots, m\}$ and $p_1 \geq p_2 \geq \dots \geq p_m$. Then the best guess of X is $\hat{X} = 1$ and the resulting probability of error is $P_e = 1 - p_1$. Fano's inequality becomes

$$H(P_e) + P_e \log(m - 1) \geq H(X).$$

The probability mass function

$$(p_1, p_2, \dots, p_m) = (1 - P_e, \frac{P_e}{m - 1}, \dots, \frac{P_e}{m - 1})$$

achieves this bound with equality.

Applications of Fano's inequality

- Prove converse in many theorems (including channel capacity)
- Compressed sensing signal model

$$y = Ax + w$$

where $A \in \mathcal{R}^{M \times d}$: projection matrix for dimension reduction.
Signal x is sparse. Want to estimate x from y .

Fano's inequality

- **Lemma 2.10.1** If X and X' are i.i.d. with entropy $H(X)$,

$$\Pr[X = X'] \geq 2^{-H(X)},$$

with equality iff X has a uniform distribution.

Fano's inequality

- **Lemma 2.10.1** If X and X' are i.i.d. with entropy $H(X)$,

$$\Pr[X = X'] \geq 2^{-H(X)},$$

with equality iff X has a uniform distribution.

- **Corollary** Let X, X' be independent with $X \sim p(x)$, $X' \sim r(x)$, $x, x' \in \mathcal{X}$. Then

$$\Pr[X = X'] \geq 2^{-H(p) - D(p||r)}$$

$$\Pr[X = X'] \geq 2^{-H(r) - D(r||p)}$$

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